

Topology First-Year Exam, May 2023, Part 1

Work the following problems and show all work. Support all statements to the best of your ability.

1. Show that every continuous map $f : S^1 \rightarrow S^2$ of the circle to the 2-sphere is null-homotopic.
2. Show that the Stone–Čech compactification $\beta(\mathbb{N})$ of the naturals is uncountable.
3. Let $X_{\geq 0}$, $Y_{\geq 0}$, and $Z_{\geq 0}$ denote the nonnegative parts of the x -axis, the y -axis, and the z -axis in $\mathbb{R}^3$ respectively. Let $A = \mathbb{R}^3 \setminus (X_{\geq 0} \cup Y_{\geq 0} \cup Z_{\geq 0})$. Compute the fundamental group $\pi_1(A)$.
4. Let $f : S^1 \rightarrow T = S^1 \times S^1$ be a map to a torus defined by the formula $f(z) = (z, -z)$. Compute the fundamental group $\pi_1(X)$ where $X = T \cup_f B^2$ is obtained from T by attaching a 2-disk along f .
5. Show that the projective plane $\mathbb{R}P^2$ is not homeomorphic to the Klein bottle K . What about the connected sum $\mathbb{R}P^2 \# \mathbb{R}P^2$?

Answer the following with complete definitions or statements or short proofs.

6. Show that S^2 is not homeomorphic to S^3 .
7. Let CX denote the cone over the n -point space $X = \{1, \dots, n\}$. Show that every continuous map $f : CX \rightarrow CX$ has a fixed point.
8. State the Borsuk–Ulam Theorem for S^2 .
9. Show that if $g : S^2 \rightarrow S^2$ is continuous and $g(x) \neq g(-x)$ for all x , then g is surjective.
10. State the Tychonoff theorem.
11. State the Urysohn Lemma.
12. State the Seifert–van Kampen Theorem. Can it be used to compute the fundamental group of S^1 ?
13. Show that for a retraction $r : X \rightarrow A$, for any $a_0 \in A$, the induced map $r_* : \pi_1(X, a_0) \rightarrow \pi_1(A, a_0)$ on the fundamental groups is surjective.
14. Give definition of a deformation retraction. Let A be a deformation retract of X . Show that the inclusion map $A \rightarrow X$ is a homotopy equivalence.
15. Is I^I separable?